

Olympiad Test Paper — Science (Class 7) — Paper 5

Chapter 10: Respiration in Organisms

Paper 5 — (progressively harder)

Subject	Science (NCERT Class 7)
Chapter	10 — Respiration in Organisms
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. the hardest traps combine two ideas at once — equating breathing with cellular respiration, reading gas-volume or breathing-rate data backwards, assuming exhaled air has zero oxygen or fresh air zero carbon dioxide, treating anaerobic respiration as energy-equal to aerobic, reversing the diaphragm/rib/pressure logic, swapping organs across organisms (gills/spiracles/tracheae/skin/stomata/lenticels), and mistaking photosynthesis for respiration in plants. Do not write on this paper — mark answers on your answer sheet.

1. A sealed jar contains 100 units of air made of 21 units oxygen and 79 units of other gases. A small grasshopper lives in it; a chemical absorbs every bit of carbon dioxide the moment it forms. After an hour the trapped air measures only 95 units in total. The shrinkage of 5 units is best explained by the fact that:

- (A) the grasshopper added 5 units of carbon dioxide that leaked out through the jar walls
- (B) the grasshopper made oxygen disappear and gave out 5 units of nitrogen instead
- (C) the chemical absorbed 5 units of the oxygen directly, not the carbon dioxide
- (D) the grasshopper used up about 5 units of oxygen in respiration, and the carbon dioxide that would have replaced it was soaked up by the chemical, so the total volume fell

2. A boy's breathing rate is recorded as 12 per minute asleep, 16 sitting, and 32 sprinting. He sprints for 30 seconds, then sits for 90 seconds. Treating each rate as steady during its period, the total number of breaths over these two minutes is closest to:

(A) 40 breaths

(B) 32 breaths

(C) 48 breaths

(D) 24 breaths

3. A student argues that exhaled air must contain no oxygen at all, 'because the body needs oxygen, so it would surely take every bit.' Mouth-to-mouth resuscitation, in which one person's exhaled breath revives another, is offered as evidence against this. The exhaled breath can revive a casualty because:

(A) exhaled air is enriched with extra oxygen made inside the rescuer's lungs

(B) the casualty does not actually need oxygen, only the warmth of the breath

(C) exhaled air still contains about 16% oxygen — less than the 21% inhaled, but more than enough to keep another person alive

(D) exhaled air contains pure carbon dioxide, which the casualty's cells convert back to oxygen

4. Two muscle cells start with the same amount of glucose. Cell P breaks its glucose down completely with oxygen; cell Q breaks its glucose only partway without oxygen. If cell Q needs to release the SAME total energy as cell P, then cell Q must:

(A) use exactly the same amount of glucose as cell P, since they began equal

(B) use much less glucose than cell P, because working without oxygen is more efficient

(C) break down far more glucose than cell P, because each glucose gives much less energy without oxygen

(D) use no glucose at all and rely on the lactic acid for its energy

5. Limewater in tube X is bubbled with breathed-out air; identical limewater in tube Y is bubbled for the same time with room air. X goes milky in 20 seconds; Y takes 8 minutes to reach the same milkiness. From the times alone, the best estimate is that exhaled air's carbon dioxide is roughly:

(A) exactly the same as room air's, since both eventually go equally milky

(B) many times more concentrated than room air's, since X reaches the same milkiness far faster

(C) slightly less than room air's, since X cleared first

(D) the only sample with any carbon dioxide, since Y is just water vapour

6. A frog is held gently underwater, where it stays calm for many minutes, then is placed in a sealed dry box of ordinary air, where it soon grows distressed and its skin dries. The single idea explaining BOTH the calm underwater AND the distress in the dry box is that:

(A) the frog breathes only through gills, which work underwater but fail in any air

(B) the frog stops respiring whenever it leaves the water, so air is never useful to it

(C) the frog can absorb oxygen through its skin only while the skin stays moist, so dry air starves it even though oxygen is present

(D) the frog's lungs make their own oxygen underwater but cannot do so in dry air

7. During the held pause at the very top of a deep breath, a person's chest muscles stay contracted. A doctor notes the pressure inside the lungs at that instant equals the outside air pressure (no air is moving). The moment the muscles relax, air flows OUT. The reason the air flows out is that:

- (A) relaxing enlarges the chest, lowering the lung pressure further below the outside
- (B) the lungs actively push the air out by squeezing it like a fist
- (C) the diaphragm flattens downward on relaxing, forcing the air upward and out
- (D) relaxing shrinks the chest, which raises the lung pressure above the outside pressure, and air moves from higher to lower pressure

8. A sealed flask of grape juice with yeast has a balloon over its neck. Over two days the balloon inflates, the juice smells of alcohol, and the liquid fizzes. A classmate insists the yeast is photosynthesising. The strongest single objection is that:

- (A) photosynthesis is impossible because yeast is too small to hold any chlorophyll-free gas
- (B) photosynthesis always makes lactic acid, and no lactic acid was smelled
- (C) photosynthesis would give off oxygen and need light and carbon dioxide, but here alcohol forms and the gas put out a flame, which fits anaerobic respiration, not photosynthesis
- (D) photosynthesis can only occur inside a balloon, not a flask

9. An earthworm in waterlogged soil surfaces after rain; a fish in the same flooded field's open puddle stays submerged and survives. A child says 'the worm prefers air-breathing and the fish prefers water-breathing, they are just different creatures.' A sharper, respiration-based account is that:

- (A) the worm switches to gills above ground while the fish switches to skin-breathing below
- (B) the worm needs no oxygen on the surface, and the fish needs none in the puddle
- (C) the worm's moist skin cannot get enough oxygen once water fills the soil's air spaces, so it surfaces, while the fish's gills still pull the small amount of oxygen dissolved in the puddle water
- (D) the worm comes up because the flooded soil has too much oxygen for its skin to bear

10. A green water-plant is sealed in a clear flask of water and watched all day and night. By day many gas bubbles collect and a glowing splint flares in them; by night far fewer bubbles form and a flame placed in them is snuffed. The single explanation for both day and night is that:

- (A) the plant photosynthesises by day and stops respiring, then respire by night and stops photosynthesising — so it only ever makes one gas at a time
- (B) the plant makes oxygen all the time, but the night oxygen simply dissolves away

(D) the plant respire all the time, but in bright light photosynthesis outpaces it and gives surplus oxygen, while in the dark only respiration's carbon dioxide remains

(A) 140 breaths
(B) 260 breaths
(C) 400 breaths
(D) 200 breaths

(A) burning and respiration both use oxygen and release carbon dioxide; the candle and mouse stop when oxygen runs low, while the slower-respiring seeds keep producing carbon dioxide

(B) only the candle releases carbon dioxide; the living things in jars 2 and 3 absorb it

(C) the seeds release oxygen while the candle and mouse release carbon dioxide

(D) carbon dioxide is produced only by things that visibly move or burn, not by seeds

(A) the lungs leaked oxygen out through the chest wall during exercise

(B) exercise raised the inhaled oxygen above 21%, so more remained to use

(C) the harder-working cells extracted a larger share of the inhaled oxygen, so less was left in the air breathed out

(D) the body made extra oxygen during exercise but breathed it straight back out

(A) anaerobic respiration and lactic acid (B) aerobic respiration and carbon dioxide
(C) photosynthesis and glucose (D) anaerobic respiration and alcohol

(A) exhalation, because pulling down shrinks the space and pushes air out
(B) inhalation, because the diaphragm flattening downward enlarges the chest and lowers the pressure so air flows in

(C) inhalation, because the diaphragm domes upward to make room for air

(D) exhalation, because the ribs swing up and out as the sheet moves down

16. A teacher says: 'A footballer pants for two minutes after the final whistle even though he is now standing still — this proves his muscles were short of oxygen during the match.' The reasoning works because the continued panting:

(A) blows away the alcohol his leg muscles produced during the match

(B) supplies extra oxygen needed to break down the lactic acid that piled up while the muscles outran their oxygen supply

(C) keeps the muscles making lactic acid faster now that he has stopped

(D) cools the lungs, which overheated from the fast match breathing

17. Trace a single carbon-dioxide molecule from the place it is made to the moment it leaves the body. The correct ordered path is:

(A) body cell → blood → alveoli → bronchi → trachea → nasal cavity → nostrils

(B) alveoli → blood → body cell → trachea → bronchi → nostrils

(C) body cell → alveoli → blood → trachea → nostrils → nasal cavity

(D) nostrils → nasal cavity → trachea → bronchi → alveoli → blood → body cell

18. Yeast is added to two identical jars of warm sugar water. Jar A is bubbled continuously with plenty of air; jar B is sealed airtight. Both produce gas, but only jar B builds up a strong alcohol smell. The best explanation is that:

(A) with plenty of air the yeast can respire aerobically and makes little alcohol, but sealed without air it respire anaerobically and produces alcohol along with carbon dioxide

(B) the sealed jar trapped the alcohol that the open jar simply let evaporate, though both made equal alcohol

(C) the air in jar A turned the sugar straight into alcohol, which then escaped

(D) jar B photosynthesised the sugar into alcohol because it was kept dark by the seal

19. On a cold morning a girl sees a white mist when she breathes OUT but nothing when she breathes IN. A friend claims this proves exhaled air contains carbon dioxide, 'which is the visible white gas.' The correct correction is that the visible mist is:

(A) carbon dioxide gas, which is white and visible in cold weather

(B) oxygen freezing into a fine white powder as it leaves the mouth

(C) nitrogen turning solid the instant it meets the cold air

(D) tiny water droplets formed as the warm, water-vapour-rich exhaled air cools and condenses in the cold

20. A cockroach and a fish are each sealed in their own container — the cockroach in air, the fish in water. Both eventually become distressed as their containers' supplies run low. Naming the structure each uses to take in oxygen, cockroach first then fish, these are:

- (A) gills, and spiracles
- (C) lenticels, and stomata

- (B) spiracles leading to lungs, and moist skin
- (D) tracheae fed through spiracles, and gills

21. A child claims, 'When I breathe faster I am respiring faster, and when I hold my breath I stop respiring.' Picking apart both halves, the accurate statement is:

- (A) both halves are right: breathing and respiration are the same process, so they start and stop together
- (B) faster breathing supplies more oxygen so cell respiration can speed up, but holding the breath does not stop respiration — the cells keep respiring, which is why carbon dioxide builds up and forces a breath
- (C) faster breathing has no effect on respiration, and holding the breath truly halts respiration in the cells
- (D) breathing faster slows respiration, while holding the breath speeds it up

22. Soaked living peas in a sealed flask beside limewater turn it milky overnight; an identical flask of boiled (killed) peas leaves its limewater clear. A student concludes 'boiling stops carbon dioxide from being made.' Strictly, the better-stated conclusion the control supports is that:

- (A) boiled peas make carbon dioxide but it cannot escape the dead seed coat
- (B) the carbon dioxide comes from the living peas respiring, since killed peas — which cannot respire — produce none
- (C) limewater only milks when something is boiled first
- (D) the live peas were photosynthesising in the dark, and boiling stopped it

23. At sea level a climber breathes 15 per minute; on a high peak, standing still, he breathes 24 per minute and still feels short of breath. He reasons the thin mountain air must hold a smaller SHARE of oxygen than sea-level air. Is his reasoning right, and why?

- (A) no — the air's share of oxygen stays about 21% everywhere; the air is just thinner, so each breath holds fewer total molecules of oxygen, forcing more breaths
- (B) yes — high air contains a much smaller percentage of oxygen than sea-level air
- (C) no — the percentage is higher up the mountain, but the cold slows the lungs
- (D) yes — the thin air has replaced its oxygen with extra carbon dioxide

24. Word equation on the board: glucose + oxygen → carbon dioxide + water + energy. A pupil asks why oxygen appears on the LEFT for animals' respiration but on the RIGHT when a green leaf is studied in sunlight. The correct reconciliation is that:

- (A) the leaf reverses respiration so completely that it never respire, only photosynthesises
- (B) oxygen is both used and made in respiration, so it can be written on either side at will

(C) animals and plants run the very same equation, so the side oxygen sits on is just a matter of preference

(D) respiration uses oxygen (so it is a raw material, on the left), while photosynthesis in light releases oxygen (a product, on the right); the leaf does both, and the equations are simply different processes

25. A girl counts 9 breaths in 30 seconds while sitting, then 9 breaths in 15 seconds just after a run. Converted to breaths per minute, her sitting and post-run rates are respectively:

(A) 9 and 9 per minute

(B) 18 and 36 per minute

(C) 30 and 15 per minute

(D) 18 and 18 per minute

26. A diagram shows a thin-walled air sac wrapped in a mesh of fine blood vessels, with oxygen arrows crossing from the sac into the blood and carbon-dioxide arrows crossing the other way. Naming the sac and explaining why its walls are thin, the answer is:

(A) a bronchus, whose thin walls store oxygen until the blood needs it

(B) an alveolus, whose thin walls make oxygen out of the air inside it

(C) the trachea, whose thin walls pump the blood past the air

(D) an alveolus, whose thin walls let oxygen and carbon dioxide pass quickly between air and blood

27. A coach says fitter people 'breathe less for the same work.' Runner A (trained) and runner B (untrained), same age, each do an identical step-test. A rises from 12 to 24 per minute; B rises from 18 to 30 per minute. Comparing the FRACTIONAL increase from each one's own resting rate, the data show:

(A) B's fractional increase was larger, because B reached a higher final rate

(B) both increased by the same fraction, since both rose by 12 breaths

(C) A doubled (a rise equal to its whole resting rate), while B rose by only two-thirds of its resting rate, so A's fractional jump was larger

(D) A's fractional increase was smaller, because trained people barely change

28. A respirometer holds a small living beetle; the coloured drop moves inward (oxygen used). The beetle is then killed and re-sealed with fresh air. The drop now stays still. The stillness after death best demonstrates that:

(A) the dead beetle releases oxygen, which exactly cancels its movement

(B) the inward movement was caused by living respiration consuming oxygen, which ceases when the beetle is no longer alive

(C) respirometers only work on warm-blooded animals, so a beetle never moved the drop

(D) the drop moved earlier only because the live beetle was wriggling, not respiring

29. Put these exhalation events in correct cause-and-effect order: (i) air flows out of the lungs; (ii) the pressure inside the lungs rises above the outside; (iii) the diaphragm domes

up and the ribs move down and in; (iv) the chest space gets smaller.

(A) iv $\rightarrow$ ii $\rightarrow$ i $\rightarrow$ iii

(B) ii $\rightarrow$ iii $\rightarrow$ iv $\rightarrow$ i

(C) iii $\rightarrow$ iv $\rightarrow$ ii $\rightarrow$ i

(D) i $\rightarrow$ ii $\rightarrow$ iv $\rightarrow$ iii

30. A leaf is sealed in a clear box with limewater at the bottom and left in BRIGHT light for several hours; the limewater stays clear. The same leaf and box are then kept in the DARK overnight; by morning the limewater is milky. The pair of results together shows that:

(A) the leaf respire only in the dark and photosynthesises only in the light, doing just one at a time

(B) the leaf respire all the time, but in light photosynthesis re-uses the carbon dioxide so little is left, while in the dark only respiration's carbon dioxide accumulates

(C) the leaf stops making carbon dioxide in light and starts making it only when dark

(D) the milky limewater in the dark proves the leaf was photosynthesising at night

31. A student lists four steps of inhalation but scrambles them: (i) pressure inside the lungs drops below the outside; (ii) air rushes into the lungs; (iii) chest space enlarges; (iv) diaphragm contracts and flattens. The correct cause-and-effect order is:

(A) iv $\rightarrow$ iii $\rightarrow$ i $\rightarrow$ ii

(B) ii $\rightarrow$ i $\rightarrow$ iii $\rightarrow$ iv

(C) iii $\rightarrow$ iv $\rightarrow$ ii $\rightarrow$ i

(D) i $\rightarrow$ iv $\rightarrow$ iii $\rightarrow$ ii

32. A child believes 'gills are like a fish's lungs, so a fish drowns in water the way we drown — by getting water in its lungs.' The most accurate correction is that a fish:

(A) has lungs like ours that fill with water, which is exactly why it must stay submerged

(B) makes its own oxygen from the water, so water can never harm it

(C) breathes the water directly into air sacs, the same way we breathe air into alveoli

(D) has no lungs; its gills take dissolved oxygen out of the water, so the fish needs water (not air) passing over them, and 'drowns' only when that water lacks oxygen

33. In a single bright sunny hour, a healthy leaf takes in 10 units of carbon dioxide for photosynthesis but its own respiration releases 3 units of carbon dioxide in the same hour. The NET carbon dioxide the leaf draws from the surrounding air that hour is:

(A) 7 units, because respiration's 3 units are re-used inside the leaf and only the shortfall is taken from the air

(B) 10 units, since respiration in light does not happen at all

(C) 13 units, because the leaf both absorbs 10 and releases 3 into the air

(D) 3 units, since only respiration is occurring in a sealed leaf

34. A burning candle is lowered first into a jar of breathed-OUT air and then into a jar of breathed-IN (room) air; it dies sooner in the exhaled-air jar but does burn briefly in both. A pupil concludes 'exhaled air has no oxygen.' The brief burning in the exhaled jar refutes this because it shows exhaled air:

- (A) still contains some oxygen (about 16%) — less than room air, which is why the flame is shorter-lived, not absent
- (B) contains pure carbon dioxide, which briefly fed the flame before smothering it
- (C) has more oxygen than room air, so the candle should have burned longer there
- (D) had no oxygen but the flame survived on its own stored heat

35. A boy's resting breathing rate is 15 per minute. After a fright his rate jumps by 40%, and after calming it settles to 20% above his original resting rate. His breathing rate just after the fright, and after calming, are respectively:

- (A) 40 and 20 per minute
- (B) 21 and 18 per minute
- (C) 21 and 12 per minute
- (D) 18 and 21 per minute

36. Three statements are offered: (1) breathing and respiration are the same thing; (2) plants stop respiring in daylight; (3) anaerobic respiration releases less energy per glucose than aerobic. Which statement is TRUE, and the other two false?

- (A) statement 1 is true; the other two are false
- (B) statement 2 is true; the other two are false
- (C) all three statements are true
- (D) statement 3 is true; breathing and respiration differ, and plants respire day and night

37. An athlete sprints, then rests. A graph shows her breathing rate staying HIGH for two minutes after she stops, then falling to normal. A classmate says 'her rate should drop the instant she stops, since she is no longer moving.' The reason the rate stays high is that:

- (A) her lungs cannot change speed quickly and lag behind by exactly two minutes
- (B) stopping makes the muscles produce even more lactic acid that must be blown off
- (C) the resting body needs more oxygen than the sprinting body did
- (D) her body still needs extra oxygen to break down the lactic acid that built up during the sprint, so fast breathing continues until that debt is paid

38. Which single comparison of breathing structures across organisms is entirely CORRECT?

- (A) a fish uses spiracles, a cockroach uses gills, an earthworm uses lungs, and a leaf uses lenticels
- (B) a fish uses lungs, a cockroach uses moist skin, an earthworm uses stomata, and a stem uses gills
- (C) a fish uses gills, a cockroach uses tracheae fed by spiracles, an earthworm uses its moist skin, and a woody stem uses lenticels
- (D) a fish uses gills, a cockroach uses lungs, an earthworm uses spiracles, and a root uses alveoli

39. A sealed jar of soaked germinating seeds, kept dark, is found after a night to have less oxygen and more carbon dioxide, and a candle lowered in goes out at once. A pupil says this proves the seeds were photosynthesising. The decisive objection is that:

- (A) photosynthesis needs light and would ADD oxygen, but here oxygen fell and carbon dioxide rose in the dark — the signature of respiration
- (B) photosynthesis was occurring, but the candle used up its oxygen first
- (C) the seeds were dead, so neither process happened and the gases changed by themselves
- (D) photosynthesis happens fastest in the dark, which is why oxygen fell

40. Glucose can be broken down two ways: with oxygen, giving carbon dioxide, water and a large amount of energy; without oxygen in muscle, giving lactic acid and a small amount of energy. A student says 'the oxygen-free route must release more energy, since it works without help from oxygen.' The correct rebuttal is:

- (A) the oxygen-free route releases LESS energy per glucose, because the glucose is only partly broken down, leaving energy locked in the lactic acid
- (B) the oxygen-free route releases more energy because lactic acid is an energy-rich fuel
- (C) both routes release the same energy because they start from the same glucose
- (D) the oxygen-free route releases more energy but produces it more slowly

41. A diver descends with one lungful of air. Her brain registers the urge to surface and breathe long before her muscles actually run out of usable energy. The substance whose build-up in her blood mainly drives that urge is:

- (A) oxygen, which her cells could not use up fast enough
- (B) lactic acid released directly by her lungs
- (C) carbon dioxide, produced by her continually respiring cells
- (D) alcohol made by her resting brain cells

42. A fish tank's water is slowly warmed. Warm water holds less dissolved oxygen than cool water. Watching the fish's gill covers, you would predict that as the water warms the fish will:

- (A) slow its gill movements, because warm water delivers oxygen more easily
- (B) pump water over its gills faster, working harder to gather enough oxygen from the now oxygen-poorer water
- (C) stop using its gills and gulp air from above the surface as a permanent switch to lungs
- (D) need no oxygen once the water is warm, so the gills stop altogether

43. A teacher draws the human air path and asks why a blockage in the trachea is far more dangerous to breathing than a blockage in a single one of the two bronchi. The best Class-7 reason is that:

- (A) the trachea is where energy is released, so cells starve the instant it is blocked
- (B) the trachea is the single main airway feeding BOTH lungs, so blocking it cuts off air to both, while one blocked bronchus still leaves the other lung supplied

- (C) the bronchi carry blood while the trachea carries air, so only the trachea matters
- (D) the trachea makes the body's oxygen, which the bronchi merely deliver

44. A pupil writes: 'In aerobic respiration glucose simply turns into energy; matter just becomes energy.' The most accurate correction within Class-7 chemistry is that:

- (A) the glucose disappears entirely and only energy is left, with no substances made
- (B) the glucose turns into oxygen, which is what carries the energy away
- (C) the energy itself is a substance that can be collected in a jar like carbon dioxide
- (D) the glucose and oxygen are turned into carbon dioxide and water, and energy is RELEASED in the process — the matter becomes new substances, it does not vanish into pure energy

45. A long-distance cyclist keeps a steady pace and breathes 20 times per minute for the whole ride. He covers 90 minutes of riding. Of his total breaths, exactly one-third occurred while going uphill. The number of breaths he took on the uphill sections is:

- (A) 1800 breaths
- (B) 300 breaths
- (C) 600 breaths
- (D) 60 breaths

46. A student claims earthworms 'come up after rain to drink the water,' citing how many appear on wet paths. A respiration teacher disagrees. The single best evidence FOR the respiration explanation (and against the drinking idea) is that:

- (A) earthworms breathe through moist skin using soil-air oxygen, and rainwater fills the soil's air spaces, leaving too little oxygen underground — so they surface to reach air
- (B) earthworms have mouths shaped only for drinking surface water after rain
- (C) earthworms store rainwater in their skin to use for breathing later
- (D) rain adds extra oxygen to deep soil, which the worms come up to escape

47. During a quiet, ordinary EXHALATION, which set correctly describes the diaphragm, the ribs, the chest volume, and the lung pressure all at once?

- (A) diaphragm flattens down, ribs move up and out, chest volume increases, lung pressure falls
- (B) diaphragm domes up, ribs move up and out, chest volume increases, lung pressure falls
- (C) diaphragm domes up, ribs move down and in, chest volume decreases, lung pressure rises above the outside
- (D) diaphragm flattens down, ribs move down and in, chest volume decreases, lung pressure rises

48. A brewer and a baker both add the same yeast to sugar. The brewer seals the vat and prizes the alcohol; the baker bakes the dough and the alcohol disappears in the oven, leaving holes from the gas. Naming the one process serving both, and the two products that explain both outcomes, the answer is:

- (A) aerobic respiration of sugar, producing only carbon dioxide and water
- (B) anaerobic respiration of sugar, producing alcohol (kept by the brewer) and carbon dioxide (the gas that makes the baker's holes)
- (C) photosynthesis by the yeast, producing oxygen and glucose
- (D) the boiling of the sugar liquid, producing alcohol vapour and steam

49. A child reasons: 'A leaf gives off oxygen, so plants give out oxygen the way animals give out carbon dioxide — plants must do the opposite of respiring.' The clearest single correction is that:

- (A) the oxygen a leaf gives off comes from photosynthesis in light, while the leaf is ALSO respiring and giving off carbon dioxide at the same time, just as animals do
- (B) plants never respire; giving off oxygen is their only gas activity
- (C) plants respire only at night, so by day they truly do the opposite of respiring
- (D) the oxygen given off is the same as breathing out, so plants breathe out oxygen

50. A respirometer shows a small mouse using up oxygen at a steady rate in the dark. The same chamber, with the mouse removed and only a green plant inside, in the dark also shows oxygen being used up. A pupil expects the plant to ADD oxygen. The result is explained by the fact that:

- (A) the plant photosynthesises faster in the dark, using up its own oxygen
- (B) plants never make oxygen, so the expectation was wrong in any light
- (C) the chamber leaked oxygen, which had nothing to do with the plant
- (D) in the dark the plant cannot photosynthesise, so only its respiration shows — and respiration uses oxygen, just as the mouse's did

51. A man at rest breathes 12 per minute; a doctor wants the total breaths over a whole day (24 hours) assuming this rate held constant. Working in steps, the daily total is:

- (A) about 720 breaths
- (B) about 288 breaths
- (C) about 17,280 breaths
- (D) about 1,440 breaths

52. A girl insists that during inhalation 'the lungs are muscles that suck the air in by themselves.' Correcting both the 'muscle' and the 'suck' ideas, the accurate account is that:

- (A) the lungs are strong muscles that contract to pull air inward like a pump
- (B) the lungs suck air in by lowering their own temperature below the outside
- (C) the lungs are not muscles; the diaphragm and rib muscles enlarge the chest, lowering the lung pressure, and air flows in from the higher outside pressure
- (D) the diaphragm pushes air upward into the lungs from the stomach

53. A student says, 'A frog is an amphibian, so as an adult it must still breathe with gills like a fish, plus a bit of lung.' The most accurate correction about an ADULT frog is that it:

- (A) breathes only through gills, exactly like a fish, all its life
- (B) breathes through spiracles and tracheae like an insect once grown

(C) stops needing oxygen once it can live on land

(D) breathes through its moist skin and its lungs, not gills, taking in oxygen across the wet skin and through the lungs

54. A teacher seals a small candle and a respiring grasshopper in two identical jars of equal air. The candle's flame dies after a short time; the grasshopper survives much longer before becoming distressed. Both eventually milk any limewater present. The best reason the grasshopper outlasts the candle is that:

(A) the grasshopper makes its own oxygen, while the candle only consumes it

(B) a flame burns oxygen very fast, while the grasshopper respire more slowly, so it takes longer to lower the oxygen enough to harm it

(C) the candle releases oxygen that the grasshopper then breathes in

(D) the grasshopper does not use oxygen at all, so the jar's air does not matter to it

55. Match each plant part to where its respiratory gas exchange mainly happens: a leaf, a woody stem, and a root. The fully correct set is:

(A) leaf — lenticels; woody stem — stomata; root — gills in the soil

(B) leaf — stomata; woody stem — lenticels; root — air spaces between the soil particles

(C) leaf — spiracles; woody stem — stomata; root — alveoli

(D) leaf — stomata; woody stem — alveoli; root — lenticels

56. A nurse times a patient: 24 breaths in 48 seconds. She must report whether this is at, above, or below a typical resting rate of about 15–18 per minute. Converting first, the patient's rate and the verdict are:

(A) 30 per minute, which is above the typical resting range

(B) 24 per minute, which is just above the resting range

(C) 12 per minute, which is below the resting range

(D) 48 per minute, which is far above the resting range

57. A pupil writes the respiration word equation as 'carbon dioxide + water + energy → glucose + oxygen' and labels it respiration. Identify the error and the correct version:

(A) the equation is correct respiration; energy belongs on the left because it is needed to start

(B) the only fix is to remove water; respiration is glucose + oxygen → carbon dioxide + energy

(C) the equation is correct but should read alcohol instead of glucose for aerobic respiration

(D) that equation is photosynthesis written backwards; respiration is glucose + oxygen → carbon dioxide + water + energy

58. A child holds his breath, counting how long until he must gasp. He notices that after a short jog he can hold it for far LESS time than when he is calm. The best respiration-based reason for the shorter hold after jogging is that:

- (A) jogging makes his cells respire faster, so carbon dioxide builds up in his blood more quickly, triggering the urge to breathe sooner
- (B) jogging fills his lungs with extra oxygen, which forces him to exhale sooner
- (C) jogging makes his cells stop respiring, so the urge comes from emptiness
- (D) jogging cools his blood, and cold blood demands air faster

59. A diagram shows air a person breathes IN labelled '21% oxygen, 0.04% carbon dioxide' and air they breathe OUT labelled '16% oxygen, 4% carbon dioxide.' A pupil claims 'the body added much more carbon dioxide than the oxygen it removed, so the total air volume must have grown.' The fairest correction is:

- (A) the body removed no oxygen, so only carbon dioxide was added and the volume doubled
- (B) the body added 4% carbon dioxide and removed nothing, so the volume rose by 4%
- (C) the oxygen removed (about 5%) and the carbon dioxide added (about 4%) are nearly equal, so the total volume changes very little — the percentages do not mean the air grew
- (D) the body removed all the carbon dioxide, leaving only oxygen in the exhaled air

60. Two sealed jars each hold a respiring snail and a strip of moist limewater paper. Jar 1 is also given a green water-weed in bright light; jar 2 has only the snail in the dark. After hours, jar 2's paper is strongly milky but jar 1's is barely changed. The single explanation is that:

- (A) in jar 1 the snail stopped respiring because the weed gave it oxygen
- (B) in jar 1 the lit weed's photosynthesis uses up the carbon dioxide the snail makes, while in jar 2 the snail's carbon dioxide simply accumulates in the dark
- (C) in jar 2 the snail photosynthesised in the dark, releasing carbon dioxide
- (D) the weed in jar 1 added carbon dioxide, which cancelled the snail's

Solutions — Science (Class 7), Chapter 10: Respiration in Organisms — Paper 5

Paper 5 — (progressively harder)

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	D	21	B	31	A	41	C	51	C
2	A	12	A	22	B	32	D	42	B	52	C
3	C	13	C	23	A	33	A	43	B	53	D
4	C	14	A	24	D	34	A	44	D	54	B
5	B	15	B	25	B	35	B	45	C	55	B
6	C	16	B	26	D	36	D	46	A	56	A
7	D	17	A	27	C	37	D	47	C	57	D
8	C	18	A	28	B	38	C	48	B	58	A
9	C	19	D	29	C	39	A	49	A	59	C
10	D	20	D	30	B	40	A	50	D	60	B

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (D) Respiration takes in oxygen and gives out carbon dioxide. Normally the volume would stay roughly the same because each oxygen used is replaced by a carbon dioxide given off. Here the carbon dioxide is absorbed as fast as it forms, so the volume drops by exactly the oxygen consumed — about 5 units. The claim that carbon dioxide leaked through the walls is wrong: the jar is sealed and the loss is the oxygen, not escaped gas. Nitrogen is not produced by respiration, so the idea of 5 units of nitrogen replacing oxygen is false. The absorbing chemical is chosen to soak up carbon dioxide, not oxygen; it does not remove the oxygen itself.

2. (A) While sprinting at 32 per minute for half a minute he takes $32 \times 0.5 = 16$ breaths. While sitting at 16 per minute for one and a half minutes he takes $16 \times 1.5 = 24$ breaths. The total is $16 + 24 = 40$ breaths. The figure of 32 wrongly counts only the sprint rate as if it lasted a whole minute. The figure of 48 wrongly uses the sprint rate for the full two minutes. The figure of 24 counts only the sitting period and forgets the sprint breaths.

3. (C) Each breath uses only some of the oxygen, dropping it from about 21% to about 16%; the leftover oxygen in exhaled air is enough to keep a casualty alive — which is the whole point of mouth-to-mouth. Lungs do not manufacture oxygen, so the idea of enriched air is false. The casualty certainly needs oxygen, not just warmth. Exhaled air is not pure carbon dioxide (it is only about 4%), and cells cannot turn carbon dioxide back into oxygen.

4. (C) Anaerobic breakdown releases only a small part of the energy locked in each glucose, because the glucose is not broken down completely. To match cell P's total energy, cell Q must consume many more glucose molecules. Using the same amount would leave Q far short of P's energy. The claim that oxygen-free respiration is more efficient reverses the truth — it is far less efficient per glucose. Lactic acid is a leftover, not a fuel the cell burns instead of glucose.

5. (B) Both samples turn limewater milky, so both contain carbon dioxide — room air is not carbon-dioxide-free. The faster a sample milkens, the more carbon dioxide it carries, and X (exhaled) does it in 20 seconds versus Y's 8 minutes, so exhaled air is many times richer. Reaching the same final milkiness does not make the concentrations equal; speed reveals the difference. X clearing first means MORE carbon dioxide, not less. Y also milkens, so it is not mere water vapour and does contain carbon dioxide.

6. (C) An adult frog takes in oxygen partly through its moist skin (and partly through lungs); underwater the skin stays wet and gas exchange continues, so it is calm. In a dry box the skin dries out and skin-breathing fails, so it is distressed despite the air. Adult frogs do not rely on gills, so 'only gills' is wrong. The frog does not stop respiring out of water — every cell keeps respiring. Lungs do not manufacture oxygen anywhere.

7. (D) Air moves only when a pressure difference exists, flowing from high pressure to low. At the held top the pressures are balanced and nothing moves. Relaxing the breathing muscles shrinks the chest, which squeezes the lung air into a smaller space and raises its pressure above the outside; air then flows out. Relaxing does not enlarge the chest — that is inhalation. The lungs are passive, not muscular fists. On relaxing the diaphragm domes UP, not flattens down; flattening is the active inhalation move.

8. (C) The products tell the story: alcohol plus a flame-extinguishing gas (carbon dioxide) are the signature of anaerobic respiration of sugar by yeast. Photosynthesis would release oxygen (which feeds a flame) and require light and carbon dioxide as a raw material — the opposite pattern. The other objections invent rules: there is no gas-holding rule about size, photosynthesis does not make lactic acid (anaerobic muscle respiration does), and photosynthesis is not confined to balloons. The decisive point is the alcohol and the flame-killing gas.

9. (C) An earthworm breathes through moist skin using the oxygen held in soil air spaces; flooding drives that air out, so it surfaces to reach oxygen. A fish's gills extract dissolved

oxygen from water, so it can stay in the puddle. The worm does not grow gills, and the fish does not switch to skin. Both still need oxygen continuously. Flooding LOWERS the available oxygen for the worm's skin, it does not add 'too much.'

10. (D) Respiration never stops in a living plant, day or night. In strong light photosynthesis runs faster than respiration, so there is a net surplus of oxygen (the glowing splint flares). In the dark photosynthesis stops but respiration continues, so the leftover gas is mainly carbon dioxide (it snuffs the flame). The plant does not stop respiring by day — that is the central misconception. Oxygen is not made at night, so 'dissolving away' is wrong, and the plant does not 'breathe out oxygen by night.'

11. (D) When a rate rises evenly, the total equals the average rate times the time. The average of 14 and 26 is $(14 + 26) \div 2 = 20$ per minute. Over 10 minutes that is $20 \times 10 = 200$ breaths. Using only the start rate ($14 \times 10 = 140$) ignores the rise. Using only the end rate ($26 \times 10 = 260$) ignores the slow start. The figure 400 doubles the correct count, perhaps by adding both rates instead of averaging.

12. (A) All three milken the limewater, so all three release carbon dioxide — burning and respiration share this. The candle and mouse consume oxygen until it falls too low to sustain a flame or an active animal, while seeds respire slowly and quietly but steadily. Seeds do not absorb the carbon dioxide; they produce it. Seeds release carbon dioxide (from respiration), not oxygen. Visible movement or fire is not required for carbon dioxide production — quiet respiring seeds prove it.

13. (C) Exercising muscles respire faster and demand more oxygen, so a bigger fraction of each breath's oxygen is taken up — leaving less (14% rather than 16%) in the exhaled air. Lungs do not leak oxygen through the chest wall. Inhaled air stays at about 21% oxygen regardless of effort, so it does not rise. The body does not manufacture oxygen; it consumes it.

14. (A) Holding one breath while working hard makes oxygen run short, so the muscles increasingly respire anaerobically. In human muscle the partial breakdown of glucose without oxygen yields lactic acid, whose build-up causes the burning ache. With abundant oxygen the muscle would respire aerobically and not build lactic acid, but the dive limits oxygen. Photosynthesis does not occur in muscle and does not make glucose for fuel here. Alcohol is the anaerobic product in yeast, not in human muscle — human muscle makes lactic acid.

15. (B) The rubber sheet is the diaphragm. Pulling it down enlarges the bottle's space, lowering the pressure, so the balloon (lung) fills — this is inhalation, in which the real diaphragm flattens and moves down. Pulling down does not shrink the space, so it is not exhalation. The diaphragm flattens (not domes up) during inhalation; doming up is exhalation. The rib movement described (up and out) accompanies inhalation, not exhalation, so that pairing is wrong too.

16. (B) During hard play oxygen lagged, so muscles respired anaerobically and built up lactic acid. After stopping, the body keeps breathing hard to take in the extra oxygen that breaks the lactic acid down — this 'oxygen debt' is exactly why panting outlasts the effort, confirming the muscles ran short of oxygen. Human muscles make lactic acid, not alcohol, so there is no alcohol to blow away. Resting reduces lactic-acid production, it does not speed it. Panting is about gas exchange, not cooling the lungs.

17. (A) Carbon dioxide is made by respiration in a body cell, passes into the blood, is carried to the alveoli, then travels out through bronchi, trachea, nasal cavity, and nostrils. The first wrong path starts at the alveoli and ends with the cell, reversing the direction of release. The second jumbles the order, sending blood after the alveoli. The last option is the inhalation path of oxygen IN, the exact reverse of carbon dioxide leaving.

18. (A) Yeast respire aerobically when oxygen is available, breaking sugar down more completely with little alcohol. Sealed off from air, it switches to anaerobic respiration, whose products are alcohol and carbon dioxide — hence the strong smell in jar B. It is not that equal alcohol formed and only evaporation differed; the sealed jar genuinely makes more alcohol because of the oxygen shortage. Air does not turn sugar straight into alcohol. Photosynthesis makes sugar and oxygen, not alcohol, and a seal is not 'darkness' that drives it.

19. (D) Exhaled air carries more water vapour than inhaled air; when this warm moist air meets cold air it cools and the vapour condenses into tiny visible droplets — the 'mist.' Carbon dioxide is a colourless gas and is not seen, so it is not the white cloud. Oxygen does not freeze into powder at everyday cold-morning temperatures. Nitrogen likewise does not solidify in ordinary cold air. The mist is condensed water, evidence of water vapour, not of carbon dioxide.

20. (D) A cockroach draws air in through openings called spiracles into a branching network of fine tubes (tracheae) that carry oxygen straight to its tissues; a fish uses gills to extract oxygen dissolved in water. Swapping them (gills for the cockroach, spiracles for the fish) reverses the organs. Insects have tracheae, not lungs, and fish do not use moist skin. Lenticels and stomata are plant structures, not animal breathing organs.

21. (B) Breathing (moving air) and respiration (releasing energy in cells) are linked but different. Breathing faster delivers more oxygen, letting cell respiration speed up — so the first half is broadly right. But holding the breath does NOT stop respiration: cells keep respiring, carbon dioxide keeps building, and that build-up forces the urge to breathe. So they are not the same process that start and stop together. Faster breathing does affect respiration (it aids it), and holding the breath does not halt cell respiration. Neither does faster breathing slow respiration.

22. (B) The control isolates the cause: living peas respire and release carbon dioxide (milky limewater), while killed peas, unable to respire, release none (clear limewater). So

the carbon dioxide is a product of living respiration. It is not that dead peas make carbon dioxide that is trapped — they make none, since respiration has ceased. Limewater milkens whenever carbon dioxide is present, not only after boiling. Photosynthesis needs light and would consume carbon dioxide, not release it, so the dark flask rules it out anyway.

23. (A) The proportion of oxygen in air is roughly 21% at all altitudes; what changes is the air's thinness — fewer molecules of everything per breath, so fewer oxygen molecules too. To gather enough oxygen the climber breathes more often. So his 'smaller share' idea is wrong; it is fewer total molecules, not a lower percentage. The percentage is not higher up the mountain, and cold is not the cause here. Thin air is not enriched with carbon dioxide replacing oxygen.

24. (D) Respiration consumes oxygen and gives out carbon dioxide, water, and energy — oxygen is a reactant on the left. Photosynthesis, the leaf's other process in light, builds glucose and releases oxygen — oxygen is a product on the right. The leaf carries out both; they are distinct reactions, not the same equation. The leaf does not stop respiring just because it photosynthesises. Oxygen is not freely placed on either side of respiration — it is strictly a reactant there. Animals do not run the photosynthesis equation, so side placement is not arbitrary.

25. (B) A rate per minute scales the count to 60 seconds. Sitting: 9 breaths in 30 s means $9 \times (60 \div 30) = 18$ per minute. Post-run: 9 breaths in 15 s means $9 \times (60 \div 15) = 36$ per minute. Reporting 9 and 9 ignores the different time windows. The pair 30 and 15 simply repeats the time intervals, not rates. The pair 18 and 18 wrongly assumes both windows were 30 seconds; the post-run window was only 15 seconds, doubling its rate.

26. (D) The gas-exchanging sac is an alveolus, and its walls are thin so the two gases can diffuse rapidly between the air inside and the blood outside. A bronchus is an air tube, not a gas-exchange sac, and walls do not 'store' oxygen. Alveoli do not manufacture oxygen — they let existing oxygen cross. The trachea is the windpipe and does not pump blood; that is the heart's job.

27. (C) Fractional increase compares the rise to each person's own resting rate. A: rise $24 - 12 = 12$ on a base of 12, i.e. $12/12 = 1$ (doubled). B: rise $30 - 18 = 12$ on a base of 18, i.e. $12/18 = \text{two-thirds}$. So A's fraction (1) is larger than B's (two-thirds), even though both rose by the same 12 breaths — a higher final rate (B) does not mean a larger fraction. Equal raw rise (12) does not mean equal fraction, because the resting bases differ. And here A's fraction is the larger, not the smaller.

28. (B) A living beetle respire, taking in oxygen, which lowers the trapped gas volume and draws the drop inward. Once dead it no longer respire, so no oxygen is consumed and the drop is still — proving the earlier movement was respiration. A dead beetle does not release oxygen. Respirometers work on cold-blooded animals too, and the drop did move

for the live beetle. Mere wriggling does not consume oxygen the way respiration does; the controlled comparison pins the cause on respiration.

29. (C) Exhalation begins with the muscles relaxing: the diaphragm domes up and ribs move down and in (iii); this shrinks the chest space (iv); the smaller space raises the lung pressure above the outside (ii); air then flows out from high to low pressure (i). The first wrong order starts with the chest shrinking before any muscle change. The second starts with the pressure rising before the cause (the chest shrinking) exists. The last begins with air already flowing out, which is the final effect, not the trigger.

30. (B) The leaf respire both day and night, always producing carbon dioxide. In bright light, photosynthesis consumes carbon dioxide as fast as (or faster than) respiration makes it, so the limewater stays clear. In the dark, photosynthesis stops but respiration continues, so carbon dioxide accumulates and milks the limewater. The leaf does not respire 'only in the dark' — it respire constantly. It does not stop making carbon dioxide in light; the carbon dioxide is just re-used. Milky limewater in the dark signals respiration, not night-time photosynthesis (which does not occur without light).

31. (A) Inhalation begins with the diaphragm contracting and flattening (iv); this enlarges the chest space (iii); the larger space lowers the lung pressure below the outside (i); air then rushes in from high to low pressure (ii). The first wrong order starts with air already flowing in, the final effect. The second puts the chest enlarging before the diaphragm acts, reversing cause and effect. The last begins with the pressure drop before its cause (the chest enlarging) has happened.

32. (D) A fish does not have lungs; its gills extract oxygen already dissolved in water. It depends on oxygen-bearing water flowing past the gills and suffers when that water runs low on oxygen (its version of 'drowning'). It does not have water-filled lungs. It does not manufacture oxygen from water — it only takes the dissolved oxygen. It does not breathe water into air sacs the way we breathe air into alveoli; the gill mechanism is different.

33. (A) Photosynthesis needs 10 units of carbon dioxide; the leaf's own respiration supplies 3 of those internally, so only $10 - 3 = 7$ units must be drawn from the outside air. The figure 10 wrongly assumes respiration stops in light — it does not. The figure 13 wrongly adds release to uptake, but the released carbon dioxide is re-used, not vented and re-absorbed. The figure 3 ignores photosynthesis, which dominates in bright light. Net uptake from the air is 7 units.

34. (A) A flame needs oxygen; the candle burns briefly in exhaled air, proving some oxygen remains (about 16%), though less than room air's 21%, so it dies sooner. Carbon dioxide does not feed a flame, so it cannot be 'pure carbon dioxide that briefly fed it.' Exhaled air has LESS oxygen than room air, not more, so the candle does not last longer there. A flame cannot persist on 'stored heat' without oxygen — its brief life is the oxygen that is present.

35. (B) A 40% rise on 15 is $15 + 0.40 \times 15 = 15 + 6 = 21$ per minute. Calming to 20% above resting is $15 + 0.20 \times 15 = 15 + 3 = 18$ per minute. So 21 then 18. The pair 40 and 20 mistakes the percentages for the rates themselves. The pair 21 and 12 wrongly subtracts 20% (12 is 20% below 15, not above). The pair 18 and 21 reverses the two results.

36. (D) Anaerobic respiration breaks glucose down only partway, releasing far less energy per glucose than the complete aerobic breakdown — so statement 3 is true. Statement 1 is false: breathing is the mechanical exchange of air, respiration is the chemical release of energy in cells. Statement 2 is false: plants respire continuously, day and night, even while also photosynthesising by day. Hence only statement 3 is true; not 1, not 2, and not all three.

37. (D) During the sprint, oxygen ran short and lactic acid accumulated; afterward the body keeps breathing fast to take in the extra oxygen needed to clear that lactic acid (the oxygen debt), so the rate stays high before settling. The lungs are not simply 'lagging' by a fixed two minutes; the duration depends on the debt. Stopping reduces lactic-acid production, it does not increase it. A resting body needs LESS oxygen than a sprinting one — the high rate is for clearing the debt, not for meeting a higher resting demand.

38. (C) Correct pairings: fish — gills; cockroach — tracheae through spiracles; earthworm — moist skin; woody stem — lenticels. The first wrong set swaps gills and spiracles between fish and cockroach, gives the earthworm lungs, and puts lenticels on a leaf (leaves use stomata). The second set gives the fish lungs, the cockroach skin, the worm stomata, and a stem gills — all wrong. The third set gives the cockroach lungs, the worm spiracles, and a root alveoli (alveoli are in human lungs, not roots).

39. (A) Photosynthesis requires light and releases oxygen; in this dark jar oxygen fell and carbon dioxide rose, which is exactly what respiration does — so the seeds were respiring, not photosynthesising. The candle was lowered in after the night and confirms low oxygen; it did not consume the oxygen beforehand. Living seeds clearly changed the gases, so they were not dead and gases did not change 'by themselves.' Photosynthesis does not happen fastest in the dark — it does not happen at all without light.

40. (A) Without oxygen, glucose is broken down only partway to lactic acid, so much of its energy stays locked in that lactic acid — far less energy is released than the complete aerobic breakdown to carbon dioxide and water. Lactic acid is a leftover the body must later clear, not a fuel that releases extra energy. Starting from the same glucose does not equalise the yields, because the degree of breakdown differs. The oxygen-free route is not 'more energy released slowly' — it is genuinely less energy per glucose.

41. (C) Cells respire continuously and keep producing carbon dioxide; with no breathing to remove it, carbon dioxide builds up in the blood and triggers the powerful urge to breathe — this happens before energy actually runs out. It is not excess oxygen; oxygen is being used up, not piling up. Lactic acid is made by muscles working anaerobically, not

'directly by the lungs,' and it is not the main breathing trigger here. Human cells do not make alcohol, so that is not the signal.

42. (B) Warm water carries less dissolved oxygen, so to meet the same oxygen need the fish must pass more water over its gills — it pumps faster. Slowing down would gather even less oxygen, the opposite of what is needed. A fish does not have lungs to switch to and cannot make air-breathing a permanent change. Warming does not abolish the fish's oxygen need; every cell keeps respiring, so the gills cannot simply stop.

43. (B) The trachea is the one shared airway that splits into the two bronchi; block it and neither lung receives air. Block just one bronchus and the other lung is still ventilated, so it is less catastrophic. The trachea is an air passage, not the site of energy release (that is inside the cells). Bronchi carry air, not blood. Neither the trachea nor anything else makes oxygen — oxygen is taken from the air, not manufactured.

44. (D) Aerobic respiration converts glucose and oxygen into carbon dioxide and water, releasing energy. The matter does not vanish — it becomes carbon dioxide and water. So glucose does not simply 'disappear into energy.' Glucose is not turned into oxygen; oxygen is a reactant used up, not a product. Energy is released, not a material substance you can bottle like carbon dioxide. The correct picture is reactants becoming products plus released energy.

45. (C) Total breaths = rate $\times$ time = $20 \times 90 = 1800$. One-third of these are uphill: $1800 \div 3 = 600$ breaths. The figure 1800 is the whole ride, not the uphill third. The figure 300 wrongly takes one-sixth (or halves the third). The figure 60 confuses minutes with breaths. The uphill total is 600.

46. (A) Earthworms exchange gases through moist skin and rely on oxygen in the air spaces between soil particles. Heavy rain floods those spaces, displacing the air, so oxygen underground drops and the worms surface to breathe — a respiration need, not thirst. They do not have special drinking mouths driving the behaviour, and they do not store rainwater as a breathing supply. Rain does not enrich deep soil with oxygen; it does the opposite by waterlogging it.

47. (C) In exhalation the breathing muscles relax: the diaphragm domes up, the ribs fall down and in, the chest volume shrinks, and the trapped air's pressure rises above the outside so air leaves. The first wrong set is the full inhalation picture (flatten, up-and-out, enlarge, pressure falls). The second mixes a doming diaphragm with rising ribs and a falling pressure — contradictory. The last pairs a flattening diaphragm (an inhalation move) with a shrinking chest, which cannot happen together.

48. (B) Both rely on yeast's anaerobic respiration of sugar, whose two products are alcohol and carbon dioxide. The brewer keeps the alcohol; the baker uses the carbon dioxide to raise the dough (the alcohol bakes off). Aerobic respiration would not produce

alcohol, so it cannot give the brewer's prize. Photosynthesis makes sugar and oxygen, not alcohol, and yeast does not photosynthesise. Boiling does not create alcohol from sugar — the alcohol comes from the yeast's respiration.

49. (A) A leaf's released oxygen is a product of photosynthesis, which runs in light. But the leaf respire constantly too, taking in oxygen and giving out carbon dioxide — the same respiration animals do. So plants do not 'do the opposite of respiring'; they respire AND photosynthesise. Plants do respire (not never), and they respire by day as well as night (not only at night). 'Breathing out oxygen' confuses the photosynthetic release of oxygen with respiration's release of carbon dioxide.

50. (D) Plants release oxygen only through photosynthesis, which needs light. In the dark the plant cannot photosynthesise, so its constant respiration is all that shows — and respiration consumes oxygen, exactly like the mouse. The plant does not photosynthesise faster in the dark (it cannot at all). Plants DO make oxygen, but only in light, so the expectation would be right in bright light, not in the dark. The controlled chamber points to respiration, not a leak.

51. (C) In one hour: $12 \times 60 = 720$ breaths. In 24 hours: $720 \times 24 = 17,280$ breaths. The figure 720 is just one hour's total, not a day's. The figure 288 confuses minutes (12×24) and forgets the per-minute scaling. The figure 1,440 is the number of minutes in a day, not breaths. The full-day total is about 17,280.

52. (C) Lungs are elastic bags, not muscles; they cannot suck. The diaphragm and rib muscles do the work, enlarging the chest so the lung pressure drops below the outside, and air then flows in from high pressure to low. So lungs do not contract like a pump. Temperature differences are not the mechanism. The diaphragm flattens DOWN (it does not push air up from the stomach) — its downward move helps enlarge the chest.

53. (D) An adult frog takes in oxygen through its moist skin and its lungs; it does not use gills as an adult. So 'gills like a fish all its life' is wrong. Frogs do not have spiracles or tracheae — those are insect structures. And a land-living frog still needs oxygen, because every cell keeps respiring. The correct adult-frog picture is moist skin plus lungs.

54. (B) Both the flame and the grasshopper consume oxygen and release carbon dioxide (both milken limewater). A flame uses oxygen rapidly and dies quickly when it falls; the grasshopper respire more slowly, so the oxygen lasts longer before distress. The grasshopper does not manufacture oxygen. The candle does not release oxygen for the grasshopper — it consumes oxygen. The grasshopper certainly uses oxygen; it is just slower, which is why it outlasts the candle.

55. (B) Leaves exchange gases mainly through stomata; woody stems through lenticels; roots take oxygen from the air trapped between soil particles. The first wrong set swaps stomata and lenticels and invents soil gills. The second uses spiracles (insect) and alveoli

(human lung) for plant parts. The third puts alveoli on a stem and lenticels on a root, mismatching the structures. Only stomata–lenticels–soil-air is correct.

56. (A) Rate per minute = $24 \times (60 \div 48) = 24 \times 1.25 = 30$ per minute, which sits above the 15–18 resting range. Reporting 24 forgets to scale the 48-second window to a full minute. Reporting 12 wrongly halves instead of scaling up. Reporting 48 confuses the time interval (48 seconds) with a breath count. The correct rate is 30 per minute — above resting.

57. (D) The pupil's equation builds glucose and oxygen while using carbon dioxide, water, and energy — that is essentially photosynthesis, the reverse of respiration. Respiration is $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} + \text{energy}$, with energy released as a product on the right. Energy is not a reactant 'needed to start' on the left in this context — it is released. Water is a genuine product of aerobic respiration, so it should not be removed. Aerobic respiration uses glucose, not alcohol (alcohol is the anaerobic yeast product).

58. (A) After jogging the cells are respiring faster and producing carbon dioxide more quickly; with the breath held, carbon dioxide rises in the blood sooner and triggers the urge to breathe earlier — hence the shorter hold. It is not extra oxygen forcing exhalation; oxygen is being used up, and carbon dioxide is the trigger. Jogging speeds respiration, it does not stop it. The trigger is rising carbon dioxide, not blood temperature.

59. (C) Oxygen falls from 21% to 16% (about 5% removed) while carbon dioxide rises from 0.04% to 4% (about 4% added); these are roughly balanced, so the overall air volume barely changes — the percentages describe composition, not a growing volume. The body does remove oxygen (about 5%), so 'removed no oxygen' is false. It does remove oxygen as it adds carbon dioxide, so 'removed nothing' is wrong. Carbon dioxide rises in exhaled air, so it is certainly not all removed.

60. (B) Both snails respire and release carbon dioxide. In jar 1 the lit weed photosynthesises, consuming that carbon dioxide as fast as it forms, so the limewater barely milks. In jar 2 there is no photosynthesis (dark, no plant), so carbon dioxide builds up and milks the paper strongly. The snail does not stop respiring just because oxygen is available. Snails do not photosynthesise. A photosynthesising weed REMOVES carbon dioxide, it does not add an offsetting amount.